

A quotient-Jacobian identity for the full $(1, -1, -2)$ -equivariant class containing the July 2026 counterexample to the Jacobian conjecture

Two propositions circulated for expert comment: is this standard?

Anonymous*

July 2026

Purpose. Let $\mathbb{C}^*$ act on $\mathbb{A}^3$ with *coordinate* weights $(1, -1, -2)$, i.e. $t \cdot (x, y, z) = (tx, t^{-1}y, t^{-2}z)$. We consider equivariant polynomial self-maps whose output components, in order, have weights $(-2, -1, 1)$; the counterexample to the Jacobian conjecture announced by Alpöge on July 20, 2026 [1], verified in [2], is of this type. We record an explicit quotient calculus for *every* polynomial map with this equivariance. The conceptual content fits in one sentence: such a map multiplies the orbit coordinate by a factor T , and the two target invariants involve the first and second powers of the orbit coordinate, so the induced map on the quotient plane acquires a Jacobian factor of exactly T^2 . The machinery is standard invariant theory; we record a class-wide quotient identity that appears not to be stated in the currently circulating factorization and cubic-model analyses (the precise, machine-verified relationship to those analyses is given under “Related formulations” below). We would be grateful to learn whether either statement is known, and if so where. The construction of the counterexample itself is due to [1] and, independently, to the derivation published in [3]; nothing in this note claims otherwise.

Setup. The invariant ring is $\mathbb{C}[x, y, z]^{\mathbb{C}^*} = \mathbb{C}[u, v]$ with $u = xy$, $v = x^2z$. The weight- w isotypic components relevant here are the $\mathbb{C}[u, v]$ -modules $A_{-2} = z\mathbb{C}[u, v] + y^2\mathbb{C}[u, v]$, $A_{-1} = y\mathbb{C}[u, v] + xz\mathbb{C}[u, v]$, $A_1 = x\mathbb{C}[u, v]$, with relations $vy^2 = u^2z$ and $vy = u(xz)$.

Lemma 1 (Normal form). *Every polynomial map $F : \mathbb{C}^3 \rightarrow \mathbb{C}^3$ equivariant with component weights $(-2, -1, 1)$ is of the form*

$$F_1 = zP(u, v) + y^2Q(u, v), \quad F_2 = yR(u, v) + xzS(u, v), \quad F_3 = xT(u, v),$$

the representation being unique up to the gauge $(P, Q) \mapsto (P + u^2H_1, Q - vH_1)$, $(R, S) \mapsto (R + vH_2, S - uH_2)$, and unique if one requires $Q \in \mathbb{C}[u]$ and $S \in \mathbb{C}[v]$. If F is a Keller map, then $T \not\equiv 0$, the linear part of F is the invertible anti-diagonal $(\alpha z, \beta y, \gamma x)$ with $-\alpha\beta\gamma = \det DF$, and $\alpha = \beta = \gamma = 1$ may be arranged by diagonal linear changes of source and target. Precisely, the equivalence used is left-right equivalence $F \mapsto D_{\text{tgt}} \circ F \circ D_{\text{src}}$ with $D_{\text{src}}, D_{\text{tgt}}$ invertible diagonal (such maps commute with the torus action, hence preserve the equivariance class);

*Pseudonymous deposit. Correspondence: minustsquared@gmail.com. A SHA-256 commitment to the authors' identity is published in `COMMITMENT.txt` in the accompanying deposit.

the numerical value of the constant $\det DF$ is not preserved — it rescales by the product of the six diagonal entries — but the Keller property is, and no determinant-one constraint is imposed. The known counterexample has normal form $P = (1 + u)^3$, $Q = (1 + u)(4 + 3u)$, $R = 1 + 12u + 9u^2 + (6 + 3u)v$, $S = 3$, $T = 2 - 3u - v$.

Proof. A monomial $x^i y^j z^k$ has weight $i - j - 2k$. For weight 1: $i = j + 2k + 1$, so the monomial is $x u^j v^k$. For weight -1 : if $j \geq 1$ it is $y u^{j-1} v^k$; if $j = 0$ then $k \geq 1$ and it is $x z u^j v^{k-1}$; both representations exist iff $j, k \geq 1$, whence the single relation $vy = u(xz)$, and the gauge as stated. In module terms: the kernel of $\mathbb{C}[u, v]^2 \rightarrow A_{-1}$, $(R, S) \mapsto yR + xzS$, is the free rank-one module generated by $(v, -u)$, which is exactly the gauge $(R, S) \mapsto (R + vH_2, S - uH_2)$. Weight -2 is identical with generators z, y^2 and relation $vy^2 = u^2z$; there the kernel of $(P, Q) \mapsto zP + y^2Q$ is generated by $(u^2, -v)$. The gauge shifts Q by $-vH_1$ and S by $-uH_2$, so the v -multiples of Q and the u -multiples of S can be removed, arranging $Q \in \mathbb{C}[u]$ and $S \in \mathbb{C}[v]$. This normal form is unique: every monomial of y^2Q with $Q \in \mathbb{C}[u]$ is free of z while every monomial of $z\mathbb{C}[u, v]$ contains z , and every monomial of xzS with $S \in \mathbb{C}[v]$ is free of y while every monomial of $y\mathbb{C}[u, v]$ contains y , so no further cancellation is possible and the residual gauge is 0. For the Keller statements: $\det DF$ is constant, hence equals its value at 0, the determinant of the linear part; the only equivariant linear monomials are z, y, x in the three slots; and $T \equiv 0$ would force $F_3 \equiv 0$. $\square$

Proposition 1 (Square-Jacobian identity). *Let F be as above and let $G : \mathbb{C}^2 \rightarrow \mathbb{C}^2$ be the induced map on the categorical quotient $\mathbb{A}^3 // \mathbb{C}^* \cong \mathbb{A}^2$, explicitly*

$$G(u, v) = (T(uR + vS), T^2(vP + u^2Q)),$$

which is the expression of the target invariants F_2F_3 and $F_1F_3^2$ in the source invariants. Then, identically in $\mathbb{C}[u, v]$,

$$\det DG = -T^2 \det DF.$$

In particular F is Keller with $\det DF = c$ iff $\det DG = -cT^2$: the quotient map of an étale equivariant map has Jacobian a nonzero constant times a perfect square. Its critical locus is the scheme $V(T^2)$, with reduced support $V(T)$; when T is nonconstant this is a divisor which G contracts to the origin (when T is a nonzero constant the critical locus is empty), and the multiplicity two is forced by the mechanism described in the opening paragraph.

Proof. On $x \neq 0$, (x, u, v) are coordinates and $dx \wedge du \wedge dv = x^3 dx \wedge dy \wedge dz$, since $du \wedge dv = (x dy + y dx) \wedge (2xz dx + x^2 dz)$ contributes only $x^3 dy \wedge dz$ after wedging with dx . On the target, with coordinates (p, q, r) and invariants $u' = qr$, $v' = pr^2$, the same computation gives $du' \wedge dv' \wedge dr = -r^3 dp \wedge dq \wedge dr$. Pulling back by F : the left side is $d(G_1 \circ \pi) \wedge d(G_2 \circ \pi) \wedge d(xT) = (\det DG) du \wedge dv \wedge (T dx + x dT) = (\det DG) T du \wedge dv \wedge dx$, because $dT \in \mathbb{C}[u, v] du + \mathbb{C}[u, v] dv$; the right side is $-(xT)^3 (\det DF) dx \wedge dy \wedge dz$. Comparing via $du \wedge dv \wedge dx = dx \wedge du \wedge dv = x^3 dx \wedge dy \wedge dz$ yields the polynomial identity $(\det DG) T x^3 = -x^3 T^3 \det DF$ on the dense open set $x \neq 0$, hence in $\mathbb{C}[x, u, v]$. If $T \equiv 0$ then $G = (0, 0)$ and both sides of the asserted identity vanish. If $T \not\equiv 0$, cancel $x^3 T$ in the integral domain $\mathbb{C}[x, u, v]$ to obtain $\det DG = -T^2 \det DF$ in $\mathbb{C}[u, v]$. $\square$

Remark 1. *Conceptually this is the transformation law of the quotient canonical form: contracting $dx \wedge dy \wedge dz$ with the Euler field $E = x\partial_x - y\partial_y - 2z\partial_z$ of the action produces the horizontal 2-form descending to $du \wedge dv$ (up to the orbit coordinate), and T is the multiplier by*

which F scales the orbit direction, entering squared through the two target invariants of weights involving F_3 and F_3^2 . We regard the abstract principle as standard; our question concerns the explicit formula and its use below.

Proposition 2 (Generic degree). *If F is dominant and $T \not\equiv 0$ (both automatic for Keller maps), then*

$$[\mathbb{C}(x, y, z) : \mathbb{C}(F_1, F_2, F_3)] = [\mathbb{C}(u, v) : \mathbb{C}(G_1, G_2)],$$

i.e. $\mu(F) = \mu(G)$. Here $[\cdot : \cdot]$ denotes field-extension degree; over $\mathbb{C}$, for a dominant map, this coincides with the cardinality of the generic fiber (and with the separable degree, characteristic zero making all extensions separable), so no ambiguity among the usual notions of “generic degree” arises.

Proof. Set $K = \mathbb{C}(u, v)$ and $L = \mathbb{C}(x, y, z)$; since $y = u/x$ and $z = v/x^2$, $L = K(x)$. Let $w = F_3 = xT(u, v)$. Since x is transcendental over K and $T \in K^\times$, $w = xT$ is transcendental over K and $x = w/T \in K(w)$, so $L = K(x) = K(w)$ is a purely transcendental extension of K . From $G_1 = F_2F_3$ and $G_2 = F_1F_3^2$ we get $F_2 = G_1/w$ and $F_1 = G_2/w^2$, hence $\mathbb{C}(F_1, F_2, F_3) = \mathbb{C}(G_1, G_2, w)$. Dominance of F gives $\text{trdeg}_{\mathbb{C}} \mathbb{C}(F_1, F_2, F_3) = 3$; as $\mathbb{C}(F_1, F_2, F_3) = \mathbb{C}(G_1, G_2, w)$ and w contributes at most one to the transcendence degree, $\text{trdeg}_{\mathbb{C}} \mathbb{C}(G_1, G_2) = 2$, i.e. G is dominant, w is transcendental over $\mathbb{C}(G_1, G_2)$, and both displayed extension degrees are finite. Adjoining the transcendental w preserves relative degree, so $[L : \mathbb{C}(F_1, F_2, F_3)] = [K(w) : \mathbb{C}(G_1, G_2)(w)] = [K : \mathbb{C}(G_1, G_2)]$. $\square$

Remark 2. *For the known counterexample, G has Jacobian $2(2-3u-v)^2$ and generic degree 3, consistent with the cubic fiber description of [2]. The degree claim is certified as follows (script `check_quotient_degree.py` and transcript `quotient_degree_check.log` in the accompanying materials): the lex Gröbner basis of $\langle G_1 - s, G_2 - t \rangle$ over the exact function field $\mathbb{Q}(s, t)$, variables $u > v$, is triangular, consisting of a degree-3 eliminant in v and one element linear in u ; the eliminant is squarefree (its gcd with its v -derivative is constant), so the generic fiber has exactly 3 distinct points and $[\mathbb{C}(u, v) : \mathbb{C}(G_1, G_2)] = 3$. Fiber-by-fiber statements over $T = 0$ and on $x = 0$, where the action degenerates, require separate treatment and are not claimed here.*

Related formulations. A perfect-square Jacobian factor appears in two independent, essentially contemporaneous analyses of the specific counterexample, and the relationship of Proposition 1 to both can now be stated exactly rather than conjecturally. The derivation summary published by Lou [3] (produced by an OpenAI Codex model that independently rediscovered the map) factors a cubic as $L \cdot Q$ with $L = xt + \beta$, $Q = \gamma t^2 + \delta t + \varepsilon$, obtains the five-dimensional coefficient-map identity $\det \partial(c_3, c_2, c_1, c_0, \rho) / \partial(x, \beta, \gamma, \delta, \varepsilon) = -\rho^2$ (its Lemma 1), and passes to the slice $c_2 = \rho = 1$ via an explicit polynomial chart (x, y, z) (its Proposition 2). That chart is precisely the normal-form coordinate system of our Lemma 1, and the following dictionary is machine-verified in exact arithmetic (script `check_lou_dictionary.py`, transcript `lou_dictionary_check.log`, in the accompanying materials): $\beta = 1 + u$, and identically $T = 2\gamma$, so the contracted divisor $V(T)$ of Proposition 1 is $\{\gamma = 0\}$, the locus where the quadratic factor drops degree — the nonproperness divisor. In particular T does *not* correspond to ρ : on the slice $\rho \equiv 1$ identically, and ρ — like the ramification divisor of the symmetric-product construction obtained by Jiang working with ChatGPT and recorded by Sawin [4], both measuring coincidence of factors — is spent by the normalization, while T

survives to the quotient. The two squares are therefore *distinct objects in one geometry*: $-\rho^2$ upstream (finite ramification, removed by the resultant slice) and $-T^2$ downstream (the orbit-direction multiplier squared, contracted by G). The binary-cubic fiber calculus of [2] is the per-map instance of the resulting degree-3 quotient geometry.

Questions for the reader. (1) Are the normal form and the identity of Proposition 1 recorded anywhere, in this or equivalent notation? (2) Does the reading of the counterexample as a quotient-plane map with perfect-square Jacobian contracting $T = 0$ appear in any current treatment? (3) An earlier draft posed here the question of whether T matches Lou’s ρ ; it is resolved negatively by the verified dictionary above ($T = 2\gamma$, while $\rho \equiv 1$ on the slice). Is the resulting two-square pattern — a coincidence square spent by normalization, an infinity square contracted on the quotient — a known phenomenon for torus quotients of étale maps? (4) We are also investigating low-degree classification questions for this equivariant class; a complete, machine-certified classification through polynomial degree four, with all certificates, solver transcripts, and reproduction materials, accompanies this note in the same deposit (see the bundle’s **README** and **HANDOFF** files for the precise statements, software versions, and what “exhaustive” means there). Comments on whether the higher-degree program is worth pursuing would be greatly valued.

Acknowledgment. The quotient formulation was found, and the exploratory symbolic computations carried out, with the assistance of Claude Fable (Anthropic). Every mathematical claim above is certified by machine-checkable artifacts in exact arithmetic — certificates, solver transcripts, and verification scripts in the accompanying deposit — independent of any AI system’s reasoning and reproducible from the supplied materials.

References

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